

BLUESTOCK MUTUAL FUND ANALYTICS PLATFORM

End-to-End Mutual Fund Data Engineering, Analytics and Investor
Intelligence System

FINAL CAPSTONE PROJECT REPORT

Submitted By

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GitHub Link

[GitHub Repository](#)

Submitted To

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Declaration

I hereby declare that the project titled "Bluestock Mutual Fund Analytics Platform" is an original work completed as part of the Bluestock FinTech Internship Program.

This project involved the design and development of a complete mutual fund analytics ecosystem comprising data ingestion, ETL processing, database management, exploratory data analysis, financial performance analytics, risk assessment, dashboard development, and recommendation generation.

All datasets, analyses, visualizations, dashboards, and conclusions presented in this report were developed during the internship period using the resources and requirements provided by Bluestock.

The work presented in this report has not been submitted previously for any academic or professional certification.

Minakshi Kaushik

Executive Summary

The objective of this project was to build a comprehensive Mutual Fund Analytics Platform capable of transforming large volumes of financial data into meaningful investment intelligence. The platform integrates data engineering, database management, financial analytics, investor behaviour analysis, risk assessment, and interactive dashboard reporting into a unified analytical framework.

The project successfully processed and analyzed more than 100,000 records spanning mutual fund Net Asset Value (NAV) history, investor transactions, portfolio holdings, Assets Under Management (AUM), SIP inflows, and benchmark performance data. A robust ETL pipeline was designed to extract, transform, validate, and load information into a centralized SQLite database while maintaining complete data integrity and consistency.

The analytical engine evaluated 40 mutual fund schemes and 5,000 unique investors using performance indicators such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), Conditional Value at Risk (CVaR), and sector concentration metrics. In addition, investor cohort analysis and SIP continuity analysis were performed to understand long-term investment behaviour.

Key findings revealed significant variation in risk exposure across the mutual fund universe. Fund 101207 recorded the highest downside risk with a 95% VaR of -2.39% and a CVaR of -3.03%, whereas Fund 100025 emerged as the lowest-risk scheme with a VaR of only -0.33%. Portfolio concentration analysis identified an average HHI of 0.204, indicating moderate diversification across the analyzed funds.

One of the most significant findings of the project was the SIP continuity analysis. Out of 1,362 eligible investors, only 30 investors maintained regular investment behaviour according to the continuity threshold, resulting in a continuity rate of 2.2%. This highlights substantial investor churn risk and emphasizes the importance of investor engagement strategies.

To support investment decision-making, a rule-based recommendation engine was developed that recommends suitable mutual funds based on investor risk appetite. Combined with the interactive dashboard and analytical modules, the platform demonstrates how data-driven approaches can support better investment decisions and operational insights within the mutual fund industry.

1. Project Overview

The mutual fund industry has witnessed rapid expansion over the past decade, driven by increasing financial literacy, digital investment platforms, and the widespread adoption of Systematic Investment Plans (SIPs). As investor participation continues to grow, financial institutions generate massive volumes of transactional, portfolio, and market data on a daily basis.

Although such data contains valuable information, extracting actionable insights from it remains a significant challenge. Investors often struggle to compare funds effectively, understand portfolio risks, evaluate fund performance against benchmarks, and identify investment opportunities aligned with their financial goals. Similarly, fund managers and financial analysts require reliable analytical systems to monitor performance, track investor behaviour, and assess portfolio diversification.

To address these challenges, this project was undertaken to develop a comprehensive Mutual Fund Analytics Platform. The platform integrates data engineering, financial analytics, risk assessment, and business intelligence capabilities into a single ecosystem capable of processing large-scale mutual fund datasets and generating actionable insights.

The system was designed not only to analyze historical fund performance but also to understand investor behaviour, identify risk patterns, measure diversification, and provide recommendation support for investment decision-making.

2. Project Objectives

The primary objectives of this project were:

- Design and implement an end-to-end ETL pipeline for mutual fund datasets.
- Build a centralized SQLite database to store processed financial data.
- Perform exploratory data analysis to identify patterns and trends.
- Develop performance analytics using financial metrics such as CAGR, Sharpe Ratio, Alpha, Beta, and Maximum Drawdown.
- Implement advanced risk analytics including VaR, CVaR, Rolling Sharpe Ratio, and portfolio concentration analysis.

- Analyze investor behaviour through cohort analysis and SIP continuity analysis.
- Develop an interactive dashboard for visualization and reporting.
- Create a recommendation engine capable of suggesting suitable mutual funds based on investor risk appetite.

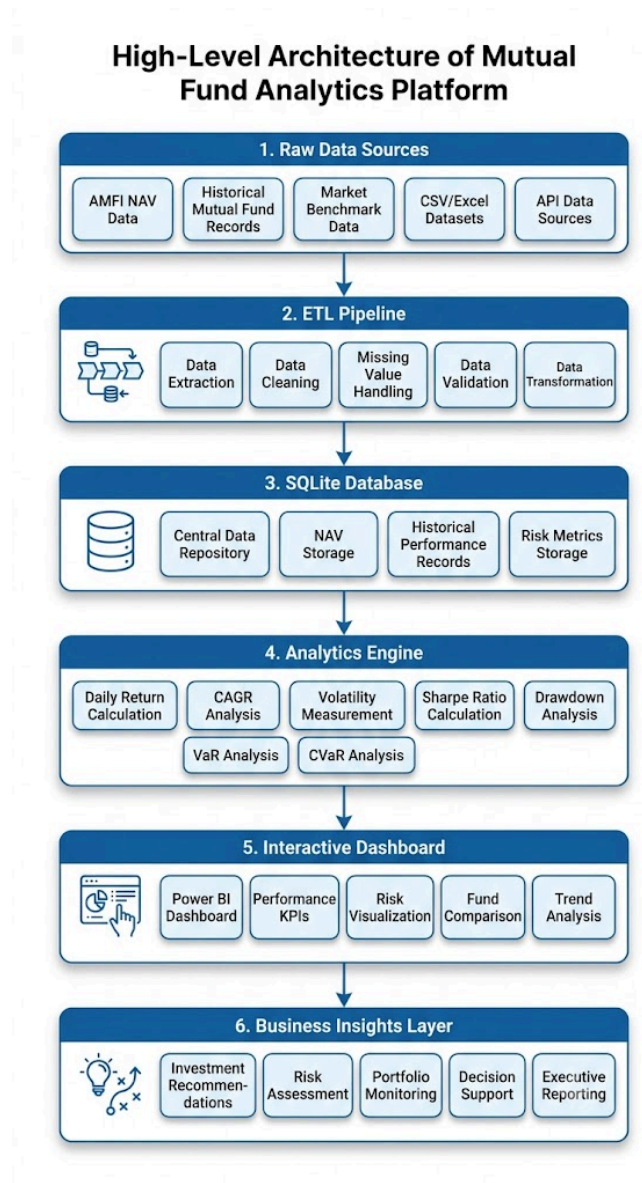


Figure 1: High-Level Architecture of the Mutual Fund Analytics Platform

3. Problem Statement

The mutual fund ecosystem generates a vast amount of data from investor transactions, fund performance records, portfolio disclosures, benchmark indices, Assets Under Management (AUM), and SIP inflows. While this information is publicly available, it is often fragmented across multiple sources and difficult to analyze in a unified manner.

Investors frequently encounter challenges when comparing funds across multiple performance dimensions. Traditional performance measures such as returns alone do not provide sufficient visibility into risk exposure, downside volatility, portfolio concentration, or benchmark-adjusted performance. As a result, investment decisions are often made without a complete understanding of risk-adjusted returns.

From an operational perspective, fund managers and analysts face similar challenges. Monitoring investor behaviour, tracking portfolio diversification, measuring churn risk, and generating timely insights require significant manual effort when analytical systems are not available.

The absence of a centralized analytical framework creates several business challenges:

- Difficulty comparing mutual funds using standardized performance metrics.
- Limited visibility into downside risk and portfolio concentration.
- Inadequate understanding of investor behaviour and SIP continuity.
- Fragmented datasets spread across multiple files and formats.
- Time-consuming reporting and manual analytical workflows.

To address these challenges, the Bluestock Mutual Fund Analytics Platform was developed as an integrated solution capable of transforming raw financial data into actionable business intelligence through automated data processing, advanced analytics, and interactive visualization.

4. Data Sources and Project Scope

The success of any analytical system depends on the quality, diversity, and reliability of its underlying data sources. This project integrates multiple datasets representing different dimensions of the mutual fund ecosystem, enabling comprehensive analysis of fund performance, investor behaviour, risk exposure, and portfolio composition.

The datasets were collected, cleaned, validated, and transformed into a unified analytical framework capable of supporting both descriptive and predictive insights.

The scope of the project extends across the complete mutual fund lifecycle, from fund-level performance tracking to investor-level transaction analysis. By combining historical NAV data, transaction records, portfolio holdings, benchmark information, and industry-level indicators, the platform provides a holistic view of mutual fund operations and performance.

The analytical coverage spans a six-year period from 2021 to 2026, providing sufficient historical depth for performance calculations, risk assessment, trend analysis, and long-term investor behaviour studies.

Dataset	Purpose
NAV History	Fund performance analysis
Investor Transactions	Behaviour analytics
Portfolio Holdings	Sector concentration analysis
Benchmark Data	Alpha & Beta calculation
AUM Data	Industry growth tracking
SIP Inflows	Investor participation trends
Fund Master Data	Scheme metadata

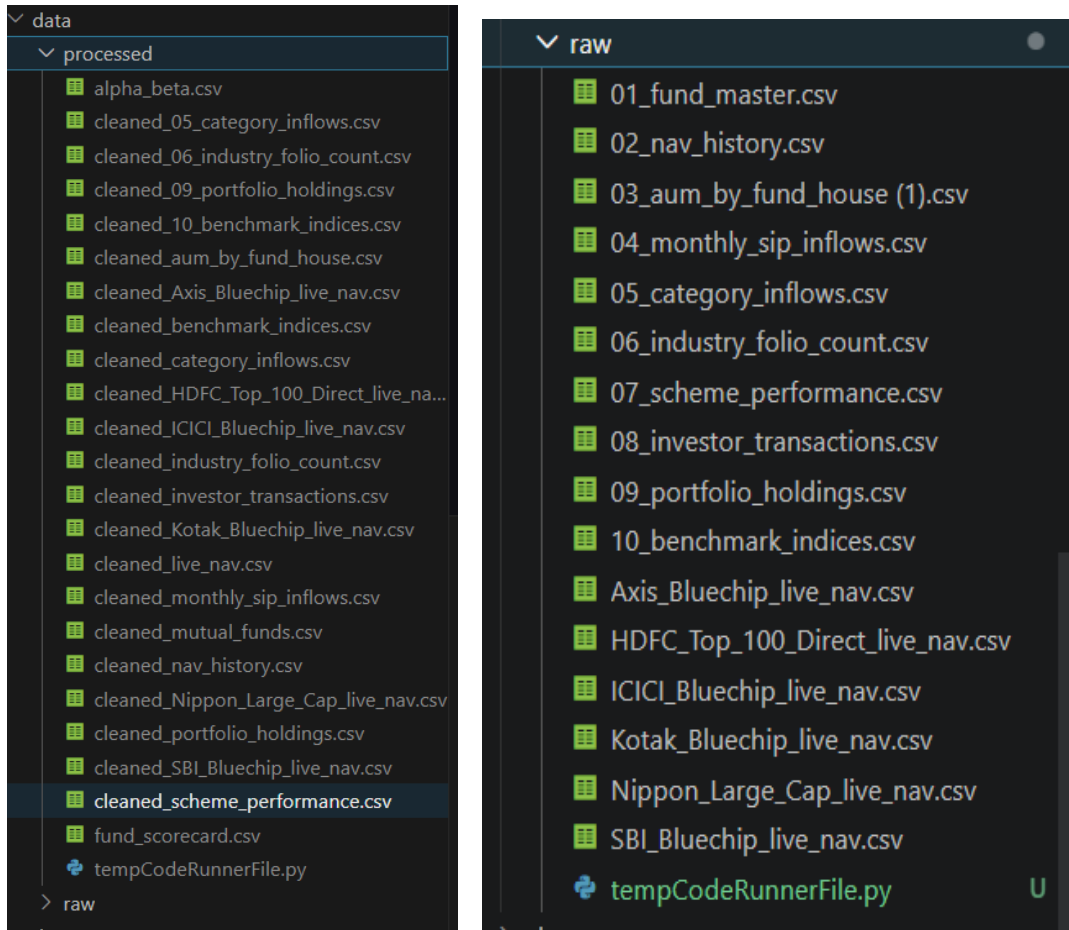


Figure 2: Raw and Processed Dataset Repository

5. Dataset Statistics and Quality Assessment

Before analytical processing, all datasets underwent validation and quality assessment to ensure reliability and consistency across the platform. The objective was to eliminate structural inconsistencies, validate identifiers, standardize formats, and confirm completeness.

The final analytical dataset consisted of more than 100,000 records distributed across multiple relational entities. Validation checks confirmed that all critical datasets were complete and suitable for downstream analytical processing.

Special attention was given to AMFI code validation, date consistency, duplicate detection, and referential integrity between transactional and master datasets. The successful completion of these checks ensured that subsequent performance and risk calculations were based on reliable information.

The quality assessment process revealed no missing values within critical analytical fields and no significant integrity violations across the database structure.

Metric	Value
Total Records Processed	100,000+
Mutual Funds Analyzed	40
Unique Investors	5,000
Portfolio Funds	34

Table	Rows
fact_nav	64,320
fact_transactions	32,778
dim_date	2,191
dim_fund	40
fact_performance	40
fact_aum	90
tracking_sip_inflows	48

Time Horizon	2021–2026
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Pipeline Health Summary

Metric	Value
Pipeline Status	SUCCESS
Data Quality	100%
Integrity Validation	PASSED
Record Loss	0%

6. ETL Pipeline Design

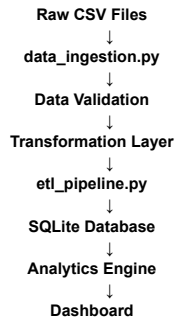
A robust Extract–Transform–Load (ETL) pipeline was developed to automate the movement of data from raw source files into the analytical database. The pipeline was designed to ensure scalability, reliability, and consistency while minimizing manual intervention.

The ETL process consists of three major stages. During extraction, raw datasets are imported from source files and validated for structural consistency. During transformation, records are standardized, cleaned, enriched, and validated. Finally, the processed datasets are loaded into a centralized SQLite database for analytical processing.

The modular architecture of the pipeline improves maintainability and allows individual components to be updated independently without affecting the overall workflow.

The ETL implementation successfully processed more than one hundred thousand records while maintaining complete referential integrity and data consistency across all entities.

ETL Pipeline Workflow :-



7. ETL Implementation Details

The ETL layer was implemented using Python and designed around modular, reusable components. This approach improved code maintainability and ensured that future datasets could be integrated with minimal modifications.

The data ingestion process was managed through the `data_ingestion.py` module, which handled file loading, schema validation, and preliminary preprocessing. Additional support for future real-time integrations was provided through the `live_nav_fetch.py` module, which enables retrieval of updated NAV information from external sources.

Transformation logic focused on improving data quality through standardization and validation procedures. Key operations included date formatting, missing value handling, duplicate removal, AMFI code verification, and feature engineering. These transformations ensured that analytical calculations were performed on clean and reliable data.

All dependencies required to execute the project were documented in the `requirements.txt` file, ensuring reproducibility across development environments.

The final stage of the ETL workflow loaded the processed datasets into the SQLite database, where they became available for analytics, dashboard reporting, and recommendation generation.

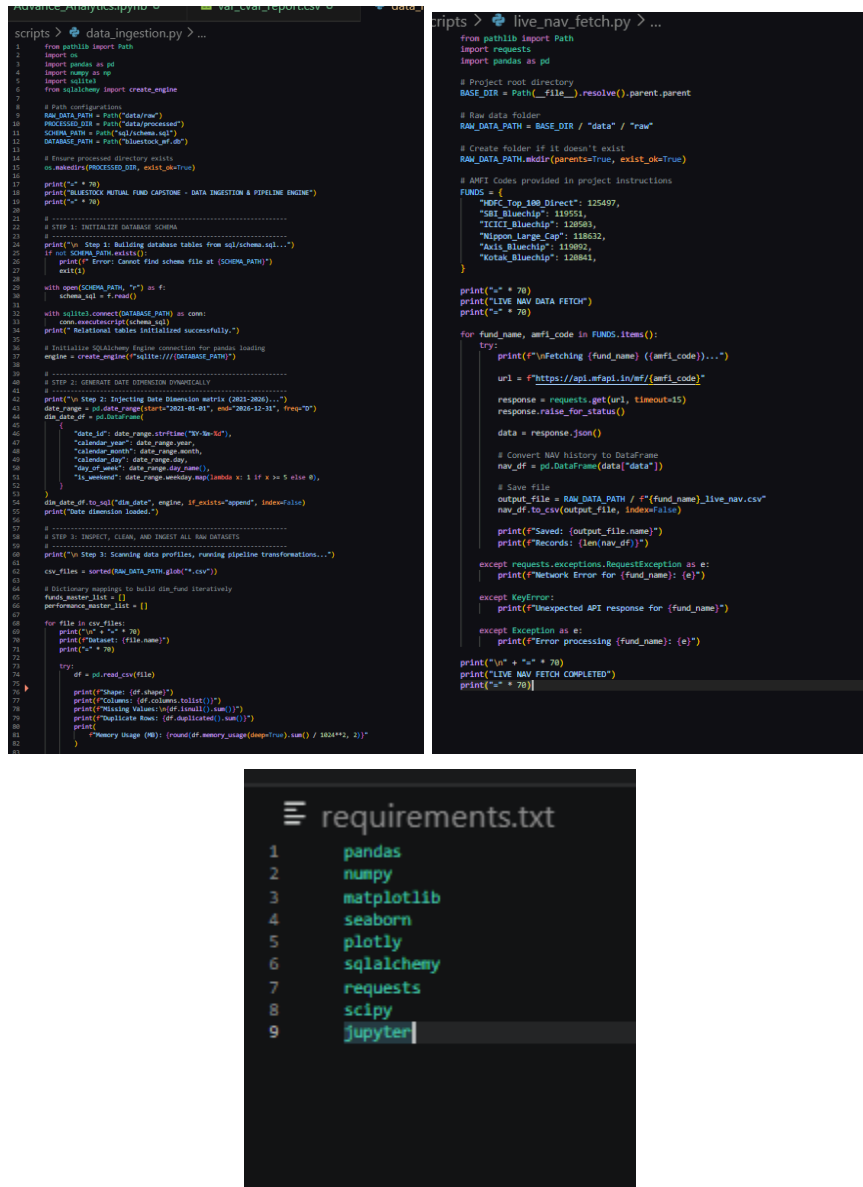


Figure 5: Core ETL Components Developed for Data Processing

8. Database Design

A centralized SQLite database was developed to serve as the foundation of the Mutual Fund Analytics Platform. The primary objective of the database layer was to organize disparate financial datasets into a structured relational model that supports efficient querying, analytics, and reporting.

The database was designed using a star-schema-inspired architecture consisting of dimension tables and fact tables. This design enables analytical operations to be performed efficiently while maintaining data consistency and reducing redundancy.

The database stores mutual fund metadata, NAV history, investor transactions, AUM records, SIP inflows, benchmark information, and calculated performance metrics. By consolidating these datasets into a single repository, the platform establishes a reliable source of truth for all analytical workflows.

The schema was implemented using SQLite due to its lightweight architecture, ease of deployment, and suitability for analytical applications. Database creation and management were supported through structured SQL scripts and documentation resources, ensuring maintainability and reproducibility.

The successful implementation of the database layer allowed the project to seamlessly transition from raw data processing to advanced financial analytics and dashboard reporting.

9. Database Tables and SQL Layer

The analytical database consists of multiple interconnected tables, each representing a distinct business entity within the mutual fund ecosystem. Together, these tables provide comprehensive coverage of fund performance, investor activity, and industry-level metrics.

The database structure was implemented using SQL scripts that automate schema creation and support analytical querying. Documentation was maintained through a detailed data dictionary describing table relationships, attributes, and business meanings.

The SQL layer enables rapid retrieval of analytical insights and serves as the foundation for dashboard visualizations and advanced financial calculations.

The following project artifacts were developed as part of the database implementation:

- `bluestock_mf.db` – Centralized analytical database
- `schema.sql` – Database schema definition
- `queries.sql` – Analytical query collection
- `data_dictionary.md` – Metadata documentation

These resources collectively ensure that the platform remains scalable, maintainable, and easy to understand for future developers and analysts.

Table Name	Description
dim_fund	Mutual fund master information
dim_date	Calendar and date dimension
fact_nav	Historical NAV records
fact_transactions	Investor transactions
fact_performance	Performance metrics
fact_aum	Assets Under Management
tracking_sip_inflows	SIP trend tracking

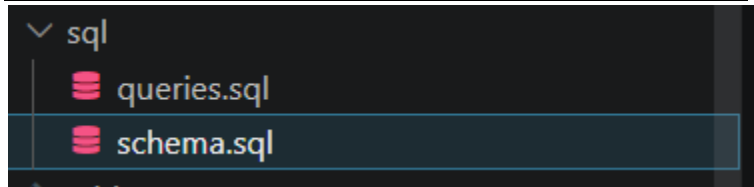


Figure 6: Database Documentation and SQL Implementation Files

10. Exploratory Data Analysis (EDA)

Before performing advanced financial calculations, an extensive Exploratory Data Analysis (EDA) phase was conducted to understand the characteristics, trends, and relationships present within the mutual fund datasets.

The objective of this phase was not only to identify data quality issues but also to uncover meaningful business patterns related to fund growth, investor participation, portfolio diversification, and market behaviour.

The EDA process was implemented through the EDA_Analysis.ipynb notebook and resulted in the generation of more than fifteen analytical visualizations. These visualizations provided valuable insights into the structure and behaviour of the mutual fund ecosystem while guiding the development of subsequent analytical modules.

The findings from this phase established the foundation for performance analytics, risk assessment, investor intelligence, and dashboard reporting.

```
notebooks > EDA_Analysis.ipynb > Import splitted
Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables

[8]
import pandas
import pandas as pd
import pandasql as ps
import pandas as pd
from pandasql import ps

[9]
SQL_QUERY = """
SELECT * FROM 'data'
WHERE year = 2018
"""

[10]
df = pd.read_sql(SQL_QUERY, conn)

[11]
df

[12]
df.head()

[13]
df.tail()

[14]
df.info()

[15]
df.describe()

[16]
df.groupby('year').mean()

[17]
df.groupby('year').std()

[18]
df.groupby('year').corr()

[19]
df.groupby('year').pct_change()

[20]
df.groupby('year').pct_change().dropna()

[21]
df.groupby('year').pct_change().dropna().head()

[22]
df.groupby('year').pct_change().dropna().tail()
```

Figure 7: Exploratory Data Analysis Workflow

10.1 Industry Growth and Investor Participation

One of the primary objectives of the exploratory analysis was to understand the growth trajectory of the mutual fund industry and evaluate investor participation trends over time.

The analysis of Assets Under Management (AUM) revealed consistent growth across the observed period, reflecting increasing investor confidence and broader participation in mutual fund products. Rising AUM levels generally indicate healthy capital inflows and stronger market engagement.

Similarly, SIP inflow trends demonstrated the growing adoption of disciplined investing practices among retail investors. Systematic Investment Plans continue to serve as one of the most effective mechanisms for long-term wealth creation, and their increasing popularity is reflected in the upward trajectory observed across the dataset.

The analysis also highlighted the role of investor demographics and geographic distribution in shaping investment behaviour. Understanding these patterns provides valuable insights into target customer segments and market expansion opportunities.

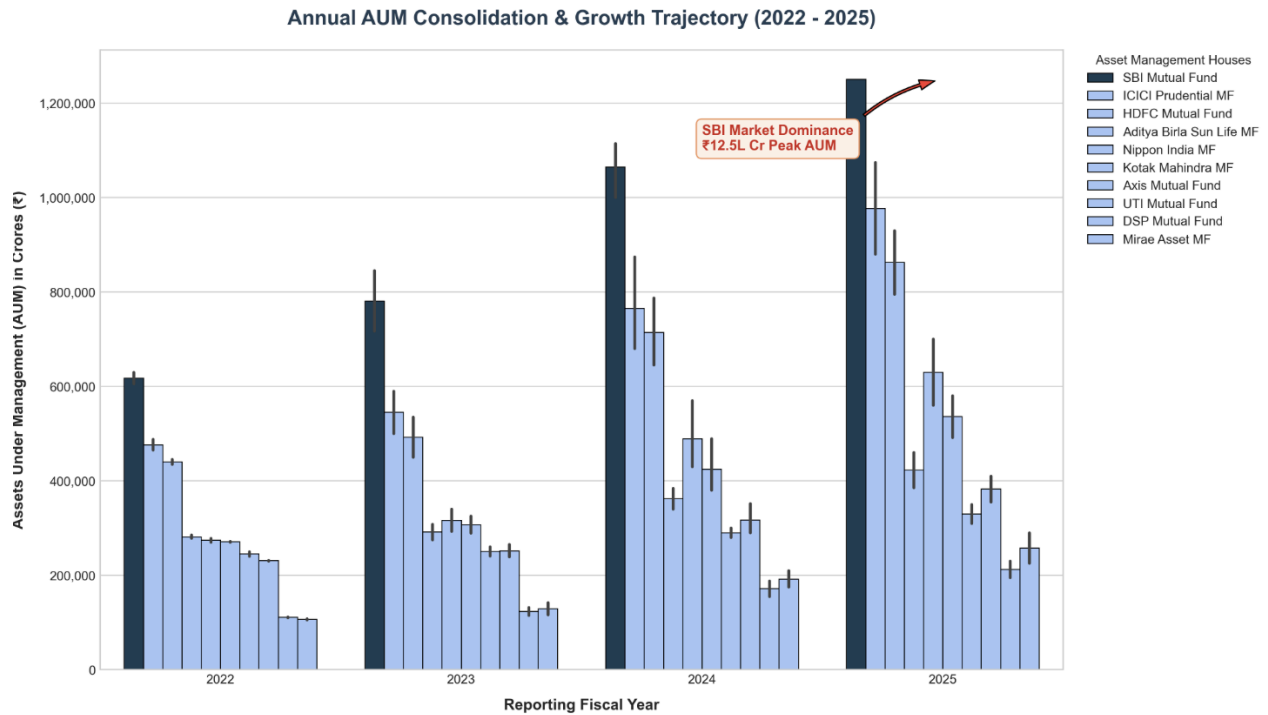


Figure 8: Assets Under Management Growth Trend

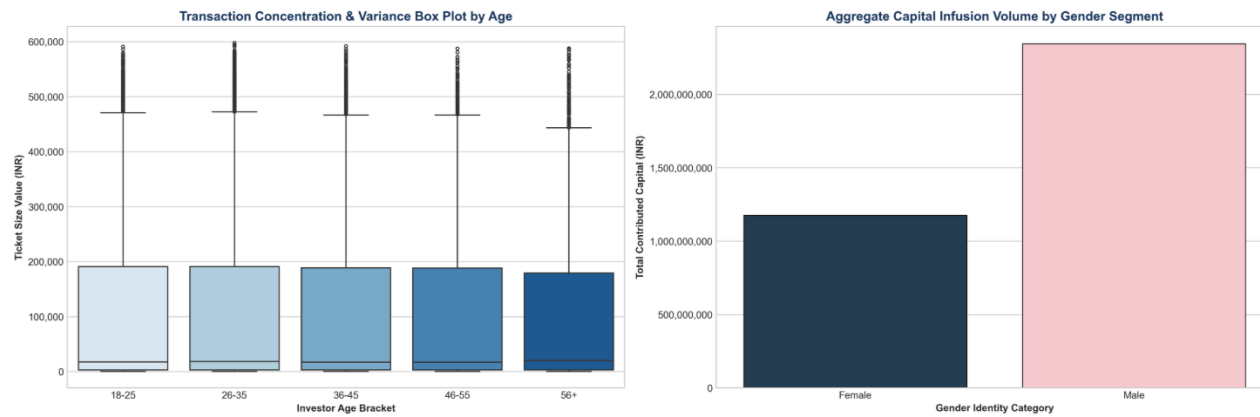


Figure 9: Investor Demographic Distribution

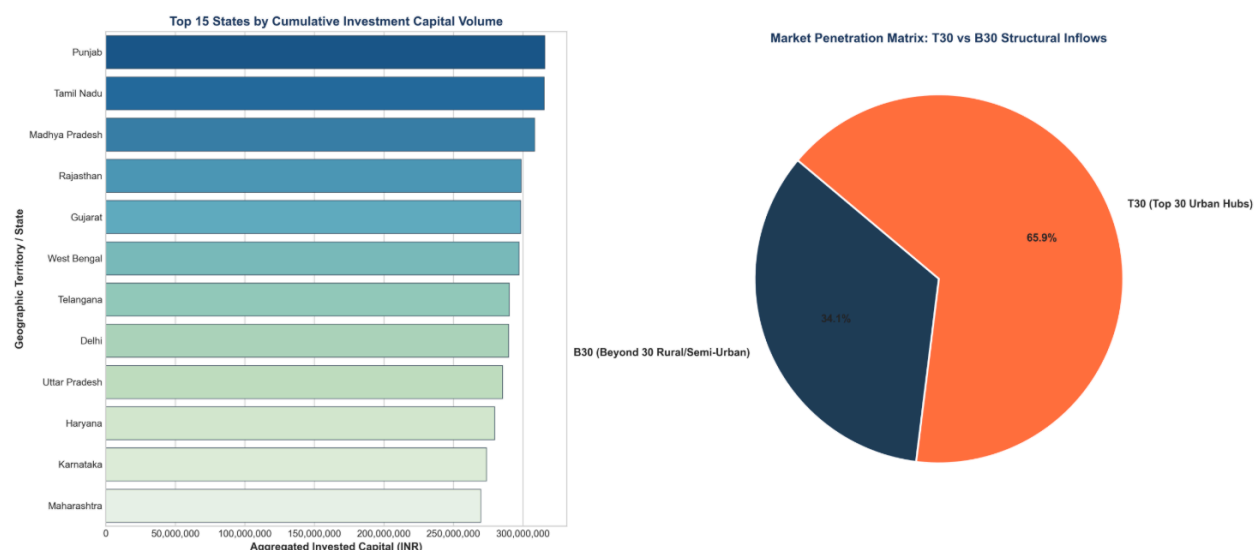


Figure 10: Geographic Distribution of Investor Activity

10.2 Diversification and Correlation Analysis

Portfolio diversification is a critical component of investment risk management. To evaluate diversification opportunities across mutual funds, a correlation analysis was performed using historical NAV returns.

The resulting correlation matrix provided insights into the extent to which different funds move together under varying market conditions. Highly correlated funds generally exhibit similar behaviour and may provide limited diversification benefits when held simultaneously. Conversely, funds with lower correlations contribute to improved risk diversification within an investment portfolio.

The analysis revealed varying degrees of correlation across the fund universe, indicating that investors have access to multiple diversification opportunities. This observation reinforces the importance of considering portfolio composition rather than evaluating funds solely on the basis of returns.

Sector allocation analysis further highlighted differences in portfolio construction strategies adopted by fund managers. Certain funds displayed broad exposure across multiple sectors, while others maintained concentrated positions in specific industries.

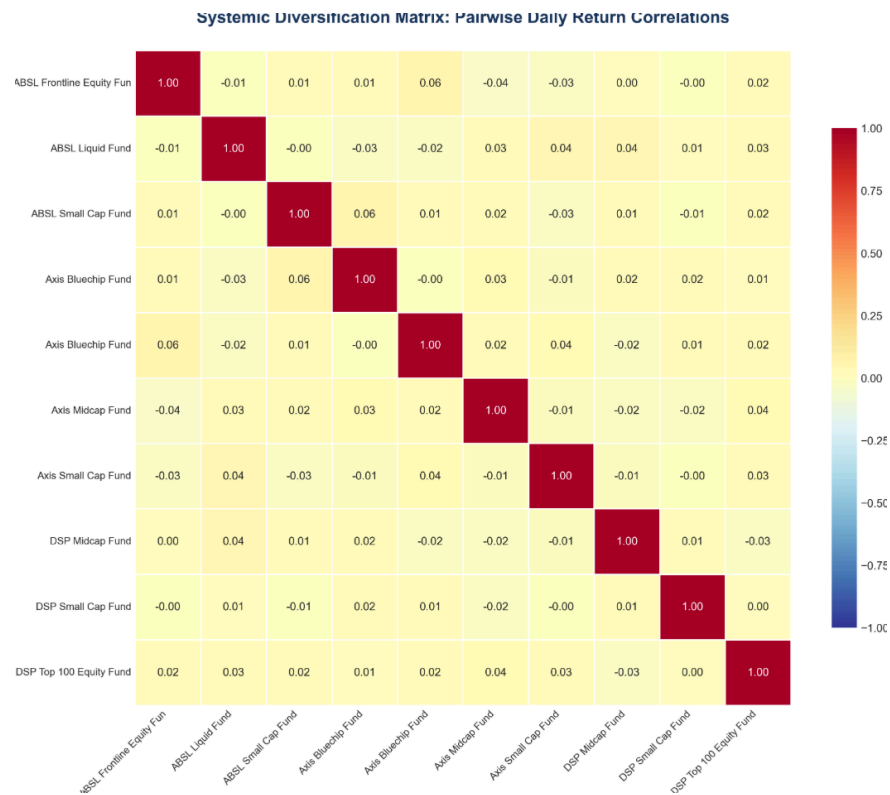


Figure 11: Mutual Fund Return Correlation Matrix

Retail Investor Commitment: Monthly SIP Inflow Vector (₹ Cr)

Quantifying systemic capital injections and all-time record allocation milestones

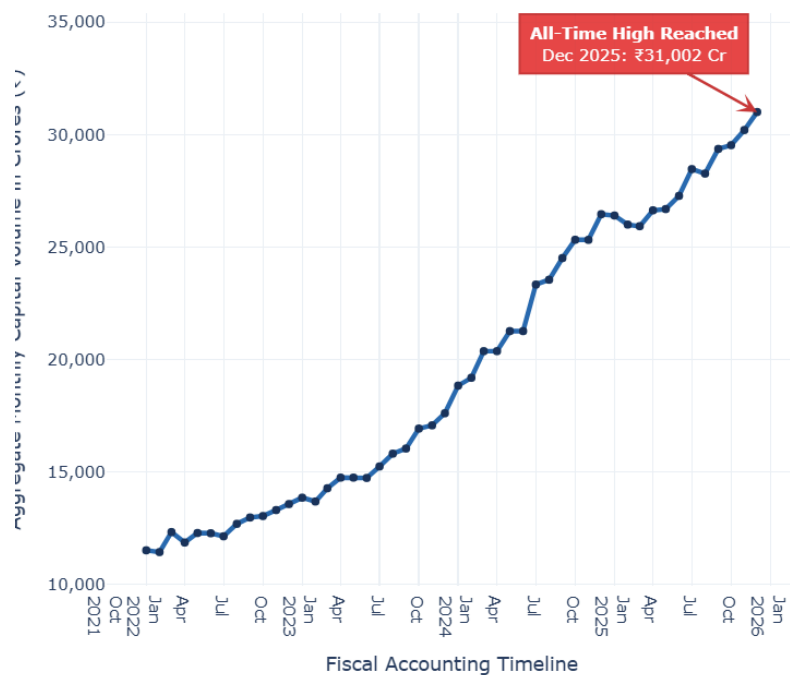


Figure 12: SIP Inflow Trend Analysis

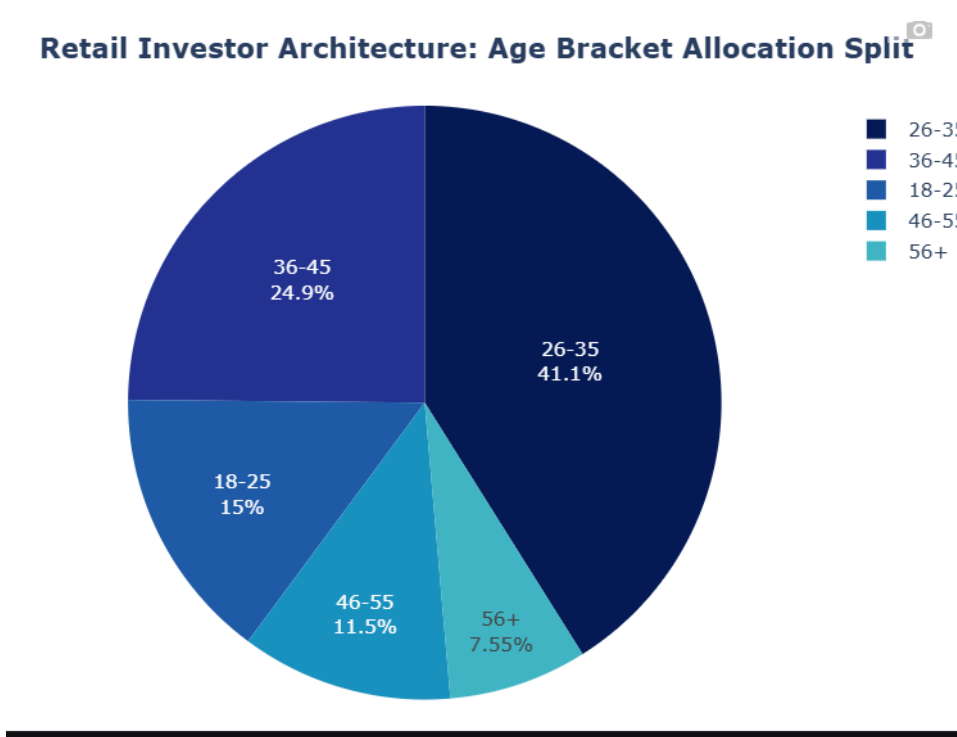


Figure 13: Investor Age Group Distribution

Key EDA Insights

- Strong growth observed in AUM and SIP participation.
- Investor activity is concentrated among specific demographic groups.
- Fund correlations vary significantly, creating diversification opportunities.
- Portfolio composition differs substantially across schemes.
- Historical patterns justify the need for deeper performance and risk analysis.

11. Performance Analytics Framework

The primary objective of the performance analytics module was to evaluate mutual fund schemes beyond absolute returns and provide a comprehensive assessment of risk-adjusted performance. To achieve this, a dedicated analytics engine was developed using Python and integrated with the centralized SQLite database.

The performance framework was designed around six widely accepted financial metrics:

- Compound Annual Growth Rate (CAGR)
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

Together, these metrics provide a balanced assessment of growth potential, risk exposure, downside protection, and benchmark-relative performance.

For each fund, daily NAV data was transformed into return series and benchmark-adjusted performance metrics were computed. The resulting values were then normalized and ranked to generate a composite Fund Score that simplifies comparison across multiple dimensions.

Fund Score Methodology

The final Fund Score was calculated using weighted rankings:

Metric	Weight
CAGR Rank	35%
Sharpe Rank	30%
Alpha Rank	25%
Drawdown Rank	10%

This scoring approach ensures that funds are rewarded not only for generating returns but also for maintaining strong risk-adjusted performance.

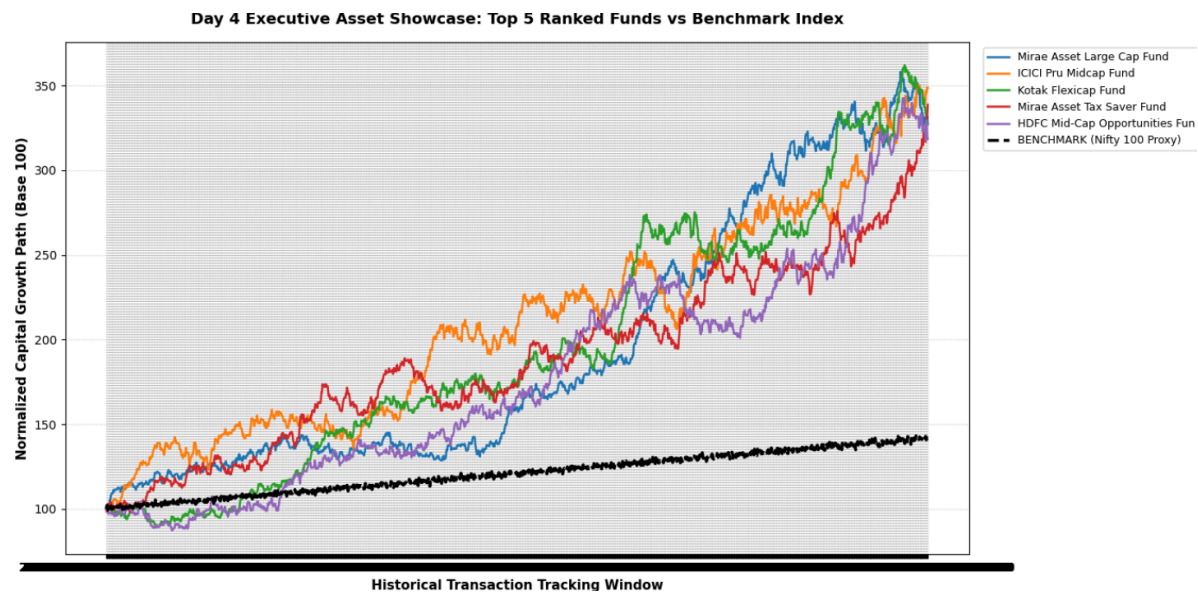


Figure 14: Top Performing Mutual Funds Compared Against Benchmark Performance

11.1 Key Performance Findings

The analysis identified significant variation in performance characteristics across the forty mutual fund schemes included in the study.

Among all evaluated funds, **Mirae Asset Large Cap Fund – Regular Growth** emerged as the strongest overall performer with a Fund Score of **91.63**.

Top Fund Performance Snapshot

Metric	Value
CAGR	20.45%
Sharpe Ratio	1.07
Sortino Ratio	1.49
Alpha	19.37%
Maximum Drawdown	-11.27%
Fund Score	91.63

The fund consistently ranked highly across multiple dimensions rather than excelling in only one area. This demonstrates the value of a multi-factor evaluation framework, as investors benefit from understanding both return generation and risk management characteristics.

The analysis also revealed that funds with strong Sharpe and Sortino ratios generally achieved higher composite rankings than funds that relied solely on return performance.

12. Advanced Risk Assessment

Traditional performance measures often fail to capture extreme downside events. To address this limitation, the project implemented advanced risk analytics using Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR).

VaR estimates the maximum expected loss under normal market conditions at a specified confidence level, while CVaR measures the average loss during the worst-case scenarios beyond the VaR threshold.

Risk Metric Results

Fund AMFI Code	VaR (95%)	CVaR (95%)
101207	-2.39%	-3.03%
100025	-0.33%	-0.46%

The results revealed a substantial difference between the highest-risk and lowest-risk schemes.

Fund **101207** emerged as the most volatile scheme within the dataset. During the worst five percent of trading days, the fund is expected to lose approximately 2.39% of its value, while losses average more than 3% during extreme downside events.

In contrast, Fund **100025** exhibited a significantly lower risk profile, demonstrating comparatively stable performance during periods of market stress.

The nearly seven-fold difference between these funds highlights the importance of incorporating downside risk measures into investment analysis.

12.1 Dynamic Risk-Adjusted Performance

To evaluate how risk-adjusted returns evolved through time, a Rolling 90-Day Sharpe Ratio analysis was performed for five representative mutual fund schemes.

Unlike static Sharpe calculations, rolling measures capture changing market conditions and provide a more realistic representation of performance consistency.

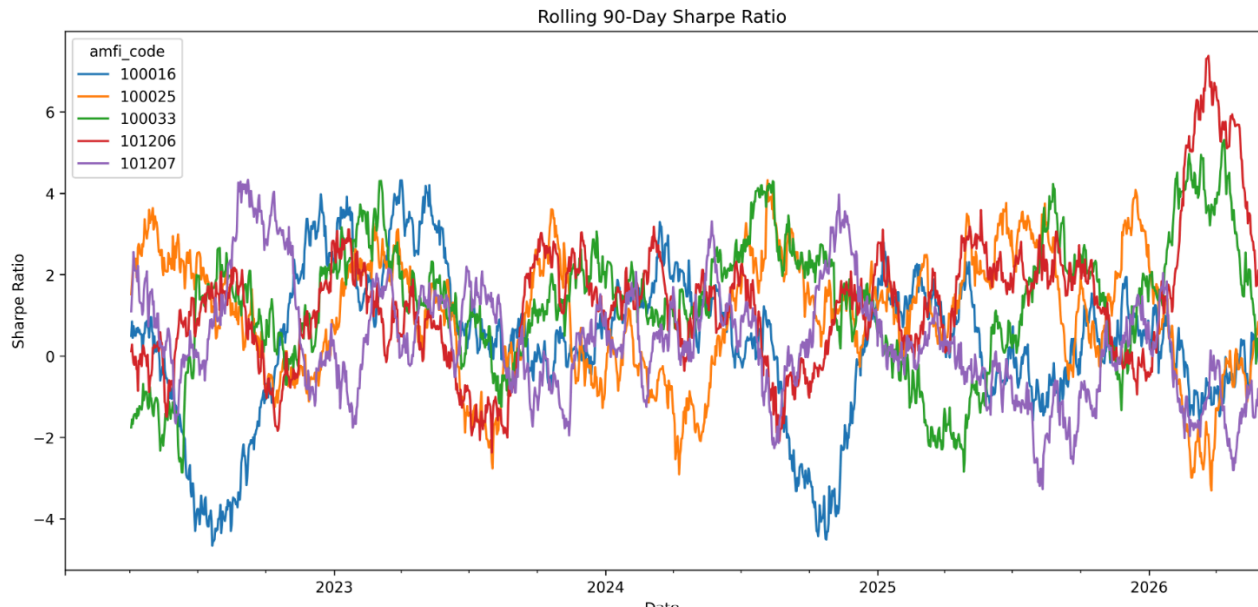


Figure 15: Rolling 90-Day Sharpe Ratios for Selected Mutual Funds

The analysis revealed substantial cyclicity across all selected funds. Fund 101206 recorded the highest Sharpe spike, approaching a value of 7.0 during early 2026. This indicates a period of unusually strong returns relative to observed volatility.

Conversely, Fund 100016 experienced the deepest negative Sharpe period, reaching approximately -4.6 during unfavorable market conditions.

These findings demonstrate that risk-adjusted performance is highly dynamic and should be evaluated across multiple market environments rather than at a single point in time.

13. Cohort and SIP Continuity Analysis

Investor behaviour plays a crucial role in determining long-term wealth creation outcomes. To better understand investment patterns, cohort analysis and SIP continuity analysis were performed using investor transaction data.

Cohort Analysis

Investors were grouped according to the year of their first recorded transaction. This enabled comparison of investment behaviour across different investor cohorts.

The analysis revealed that the 2024 cohort dominated total investment activity.

Cohort	Total Investment
2024	₹349.1 Cr
2025	₹3.05 Cr

Average SIP amounts also increased modestly between cohorts, indicating improving investor participation and confidence.

The preferred fund for the 2024 cohort was AMFI Code **148568**, while the 2025 cohort demonstrated stronger preference for Fund **119599**.

SIP Continuity Analysis

To assess investor discipline, SIP continuity was evaluated for investors with at least six SIP transactions.

Investors with an average transaction gap exceeding 35 days were classified as "At Risk."

The findings revealed a significant challenge:

- Eligible Investors: 1,362
- Active Investors: 30
- Continuity Rate: 2.2%

The low continuity rate suggests irregular investment behaviour and highlights potential opportunities for investor engagement and retention strategies.

14. Portfolio Diversification and Recommendation System

HHI Concentration Analysis

Portfolio diversification was evaluated using the Herfindahl-Hirschman Index (HHI), calculated from sector allocation weights.

The average HHI across analyzed portfolios was approximately **0.204**, indicating moderate diversification.

Key findings include:

- Most Concentrated Fund: AMFI 119092 (HHI = 0.297)
- Most Diversified Fund: AMFI 102886 (HHI = 0.124)
- Approximately five funds exceeded the 0.25 concentration threshold.

These results indicate that while most portfolios maintain reasonable diversification, several schemes carry elevated sector-specific risk.

Recommendation Engine

A rule-based recommendation engine was developed to provide fund suggestions based on investor risk appetite.

The system categorizes investors into:

- Low Risk
- Moderate Risk
- High Risk

For each category, the engine recommends the three highest-ranked funds based on Sharpe Ratio within the corresponding risk group.

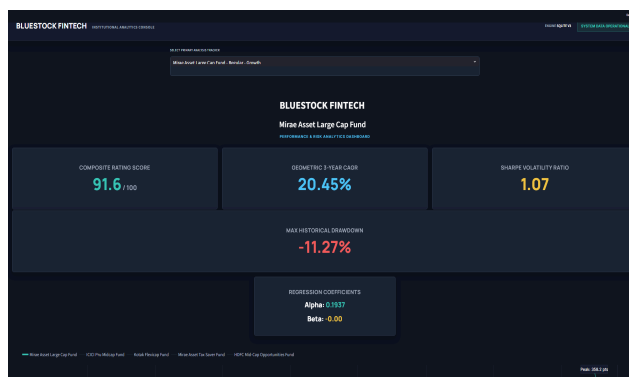
This functionality demonstrates how analytical outputs can be transformed into actionable investment recommendations.

15. Interactive Dashboard Development

An interactive business intelligence dashboard was developed using Power BI to provide a user-friendly interface for exploring mutual fund performance and investor behaviour.

The dashboard integrates multiple analytical modules including:

- Industry Overview
- Fund Performance Tracking
- Investor Analytics
- SIP & Market Trends



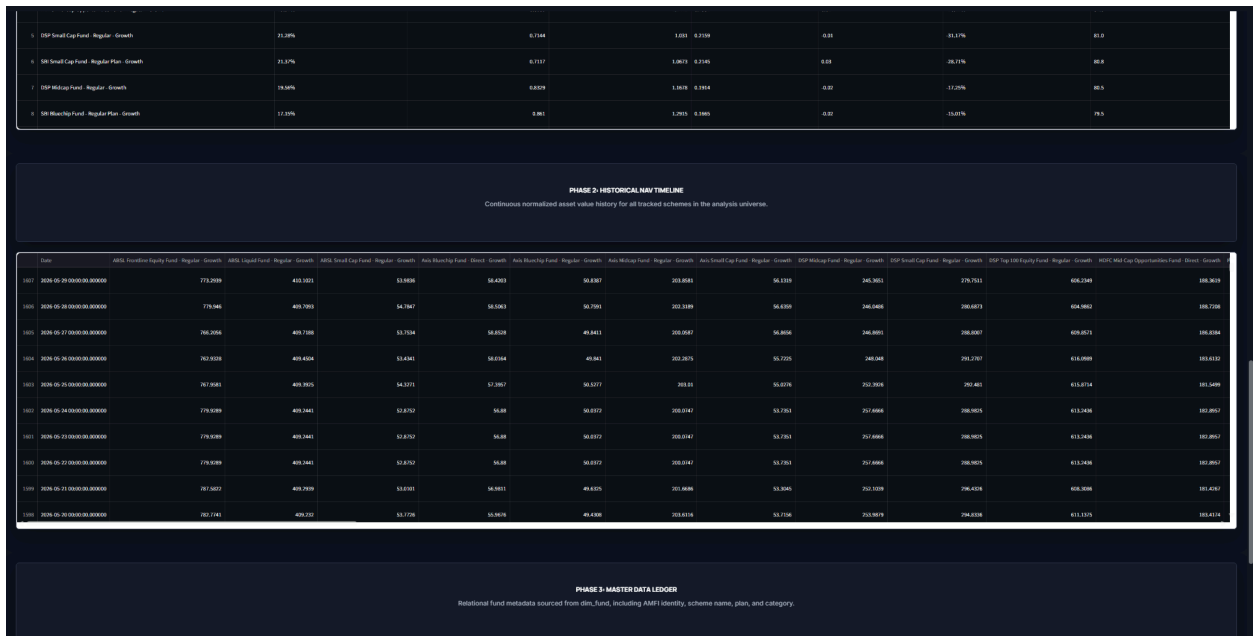
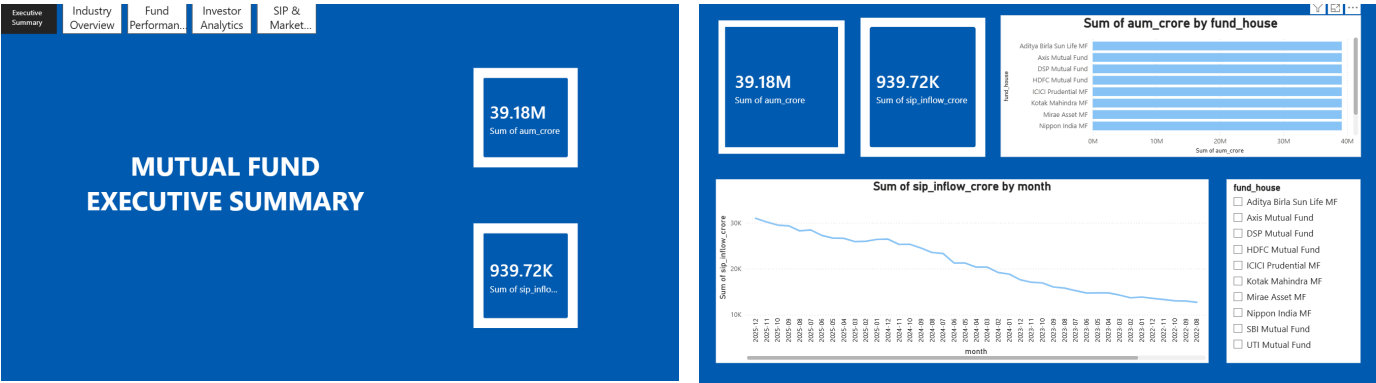


Figure 16: Dashboard Overview Page

16. Power BI Dashboard



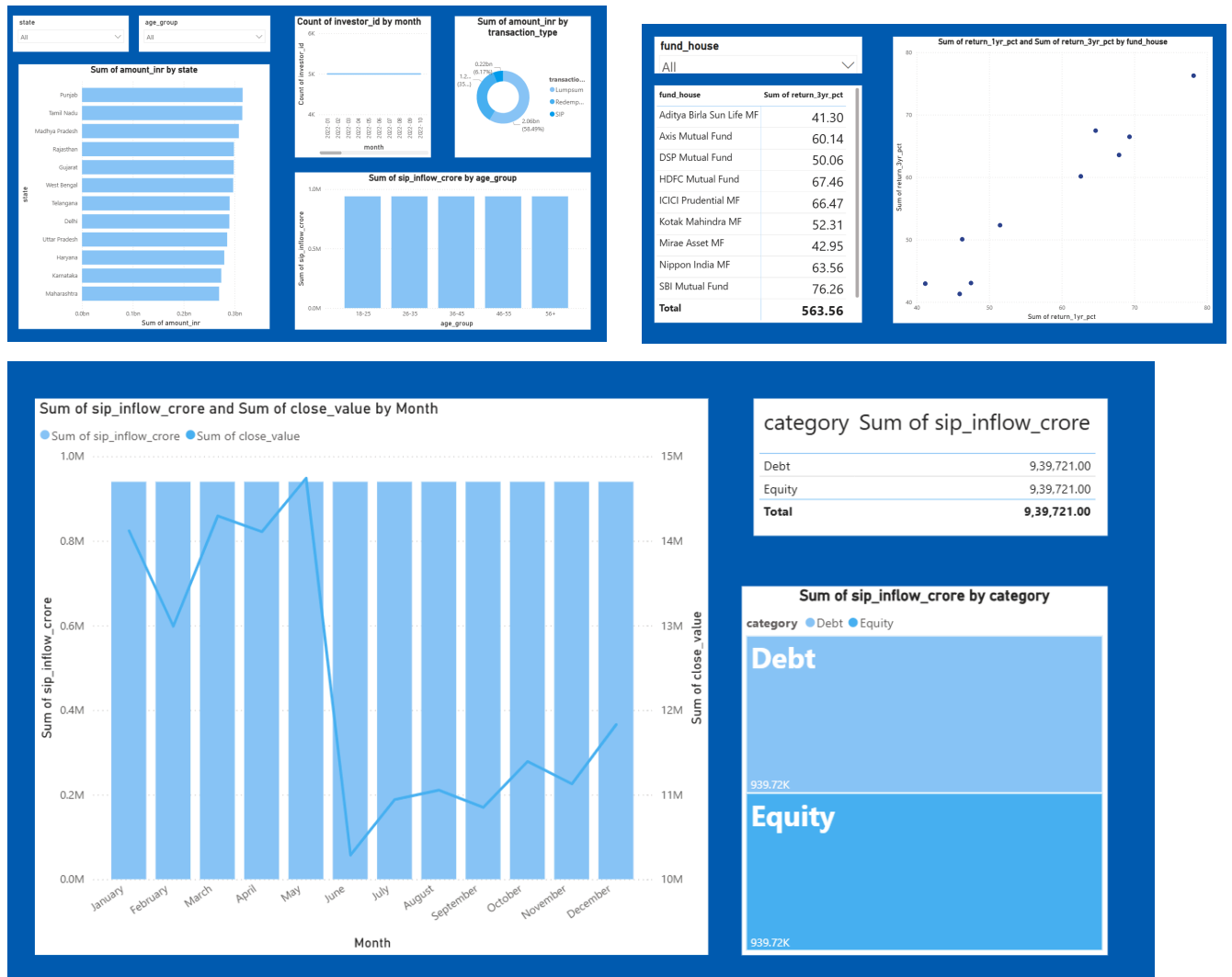


Figure 17: Power BI Dashboard Overview Page

17. Key Achievements

The project successfully:

- Processed over 100,000 records.
- Evaluated 40 mutual fund schemes.
- Analyzed 5,000 investors.
- Built an automated ETL pipeline.
- Developed advanced risk analytics.
- Created a recommendation engine.
- Designed an interactive dashboard.

- Delivered a complete end-to-end analytics ecosystem.

18. Limitations

Despite achieving its objectives, the project has several limitations:

- Analysis is based on historical data.
- Recommendation engine uses rule-based logic.
- Real-time market feeds were not integrated.
- External macroeconomic variables were not included.

19. Future Scope

Future enhancements may include:

- Machine Learning-based fund recommendations.
- Real-time NAV integration.
- Predictive risk forecasting.
- Cloud deployment.
- Investor churn prediction models.
- Personalized portfolio optimization.

20. Conclusion

The Bluestock Mutual Fund Analytics Platform successfully demonstrates the integration of data engineering, financial analytics, risk assessment, and business intelligence within a unified framework. By transforming large-scale mutual fund datasets into actionable insights, the platform enables investors, analysts, and fund managers to make more informed decisions.

The project highlights the value of data-driven investing and provides a scalable foundation for future financial analytics applications.